

CHAPTER 1

Introduction

The mismatch between community-level data gathering and local legal verification is a major barrier to the timely provision of social welfare benefits to elderly residents of Pasig.

Barangay OSCA Senior Help Desk acts only as a physical collection point for applications of Senior Citizen IDs, pensions, and burial assistance in each barangay, without the capability of instant verification of requirements or tracking document transit. This paper presents a centralized system that, through merging Rule-Based Logic and a Finite State Machine (FSM), not only connects the Barangay and the Main OSCA Office but is also capable of being used remotely by barangay personnel. The proposed system leverages automated and legally compliant guards rules to instantly audit the applicant data, allow real-time proxy-token workflows for bedridden seniors, and systematically move application states from initial intake to final LGU approval, thereby eliminating the time-consuming administrative delays and backlogs.

The Problem and its Background

As the world's population ages at a rapid pace, governments face huge challenges in helping senior citizens through social welfare programs that are both effective and easily accessible. Use of manual processing for the seniors' registration, verification and grant distribution in many developing countries, including the Philippines causes delay in administration, errors in documentation, and decreased accessibility In particular to the physically challenged people. (United Nations 2023 as cited in related digital strategy studies). At the local level these problems are Mainly visible where the local data collection by the community is often not connected to the central legal verification and approval mechanisms.Recent research

based on data indicates that senior welfare administration still has its share of problems and that going digital is one of the solutions. Research indicates that manual filing and storing of records cause accumulation of pending cases, whereas the use of electronic records management systems, rule-based automation, and workflow modeling A lot lead to efficiency, compliance, and service delivery improvement (Libadia & Botangen, 2025; Moon, 2025). With the Philippines, with the launch of the digital National Senior Citizens ID rolled out together with the eGovPH Super App, the bureaucratic red tape has been reduced and elderly have better access to benefit programs (OpenGov Asia, 2025; Philippine News Agency, 2026).

Pasig City's Office of Senior Citizens Affairs (OSCA) and barangay-level help desks operate in local government social welfare sector. Besides other vital services, issuing Senior Citizen ID, processing social pensions, assisting in burial applications, maintaining profiles, and coordination with national agencies such as Department of Social Welfare and Development (DSWD) are regularly done by these units. Normally, intake at the barangay is decentralized, after which verification and approval are centralized; Though, manual workflows often lead to slowdown (Pasig City OSCA, n.d.; Department of Social Welfare and Development, 2025).

The proposed Centralized Profiling and Record Authentication System using Rule-Based Logic and a Finite State Machine (FSM) will help automate compliance, provide workflows through proxy-token for bedridden seniors, and enable systematic changes of state from intake to final approval. The service delivery is expected to improve through faster access to senior citizens and their families, less work for barangay personnel, better data accuracy and monitoring for OSCA administrators, and even the entire LGU will benefit via improved transparency and efficient use of resources. Being able to use these technologies will help

the government to develop sustainable, inclusive governance considering the demographic changes. This is why we decided to do this research to connect the local and the central processes in Pasig City and also be in line with the national digital transformation programs (Libadia & Botangen, 2025; Department of Social Welfare and Development, 2025).

Review of Related Literature

Policy Frameworks and Digital Identity Systems

Pasig City has made significant moves to foster the welfare of its senior citizens via transparent policies. The Senior Citizens' Code of 2021 mandates the city to install various programs for seniors. The scope of these programs revolves around the registration, benefits, and daily services. The main execution of these activities has been entrusted to the Office of Senior Citizens Affairs (OSCA). These efforts will enable the city to speed up and manage the elderly people's delivery of services more efficiently (Pasig City Government, 2021).

Operating manuals of OSCA reveal the process of how these services are rendered in real-life scenarios. A senior citizen must present appropriate documents when seeking a senior ID or financial aid. The staff will review the details and carry out record updates. Monetary support is usually distributed to centenarians and other eligible seniors at set intervals. This enables the city to maintain a complete and accurate record of all its seniors (Pasig City Government, n.d.-a).

To fulfill local needs, National systems are also in place and these can help. Through determining the authenticity of IDs at different government offices, the PhilSys can provide a

trustworthy means of identity verification. That means, it limits errors and strengthens the security of data sharing. At the same time, this system is already capable to support a wide variety of public service use cases (Philippine Statistics Authority, 2020). Also, the recent launch of the digital National Senior Citizens ID is through the eGovPH app. Older adults will be able to enjoy discounts and services by simply showing their digital IDs without the need for old-fashioned cards. In this way, things are becoming more convenient for the elderly Filipinos (National Commission of Senior Citizens & Department of Information and Communications Technology, 2025).

In effect, these policies combine to foster a very conducive setting for profiling systems. Firstly, they bring together local actions and national technologies. Secondly, the outcome is a firm basis for assisting senior citizens.

Information Systems in Local Governance

Local government units must have reliable information systems to carry out their policies effectively. For instance, in Pasig City, the Management Information Systems Office is devoted to citizen data related digital solutions. Besides increasing the departments' level of coordination, these digital infrastructures can also be used by OSCA for the management and ease of availability of senior records (Pasig City Government, n.d.-b).

Studies on digital transformation in local governments highlight numerous benefits. They help in minimizing the use of paper documents. Besides that, they make the time for application processing much shorter. Officers However have an enhanced capability to monitor services and identify issues quicker than before. Seniors who normally have a tough

time physically getting to offices stand to be really benefited in this aspect (Understanding local government digital technology adoption, 2023).

Even more extensive work point out the significance of digital infrastructure. Proper systems result in greater equity in social welfare program delivery. On top of that they enable a government to make forecasting for population aging more accurate. For the Philippines, this may imply a greater efficiency in resource utilization (OECD, 2026).

In general, information systems be a link between polices and the delivery of services on a day-to-day basis. They render government operations more up-to-date and more oriented towards the needs of the citizens. This is In particular crucial when it comes to the support of elderly persons who are spread out in different cities and municipalities.

Computational Modeling Techniques

Finite state machines (FSMs) are effective methods that help visualize the processes of a system in a simple and straightforward manner. Through FSMs, one can perfectly understand the flow or progression of the system from one point to the next. This is Of course a major benefit when applying a process like senior registration or benefit approval. Besides, the laid-out format greatly makes easier testing and debugging the system (Validation of DNAFIDs model through finite state machine, 2021).

Rule-based logic is a great way for the system to make decision independently. It throws a glance at the situation using specific if-then conditions leading to an action,

which are derived by a human expert. Fuzzy logic-based systems are better when information is incomplete or inexact. This is the situation when working with data relating to a senior's health or family (Personalised serious games and gamification in healthcare, 2024).

The fuzzy state diagram method is a technique for managing uncertain situations. It gives the system flexibility and robustness by operating with incomplete or changing information. The outlined methods can be the very ones that Espanillas can use in their OSCA operations (State diagram of the fuzzy finite state machine, n.d.).

Such techniques in modeling are also compatible with the implementation of secure and flexible profiling systems. They are quite in line with the requirements of the local government. Besides, they are instrumental in building resources that are dependable and at the same time, conducive to updates.

Advanced Analytics and Ethical Considerations

Machine Learning and data mining are capable of delineating senior citizens profiles in substantial ways. They are essentially the sciences of data through which vast data sets are surveyed aiming to unveiling hidden useful patterns. Besides detection of health risks, they may point to seniors who, for one reason or the other, would need additional support. As a result, seniors receive services in a timely fashion and ones that mainly suit their needs (Machine learning prediction of progression to dialysis, 2026).

Common topics of mental health in senior citizens are better studied thanks to these techniques. Data analyses are pivotal to knowing which problems occur the most among elderly persons. The garnered information is instrumental to designing community activities and planning individual care (Mental health care for older adults, 2022).

Protecting privacy with the utilization of advanced tools is a must. Privacy of data can be better protected with edge-AI solutions by processing data locally. When it comes to handling sensitive information of senior citizens, such method mitigates the risks (ElderMate, n.d.).

Ethical principles are Really a must to direct the development of such systems. They are the ones that guarantee that the technology is used in a manner consistent with fairness and devoid of any form of prejudices. Most importantly, guidelines call for transparency and respect of human rights to be upheld whenever public services are involved (Universal guidelines for AI, n.d.).

Synthesis

Senior citizen profiling system needs, drawing from various literature, is a topic that has been greatly explored. Basic rules and identity instruments are offered by the policies of Pasig City and the national programs like PhilSys. Working services are implemented through the transformation of rules to data management and performance by information systems. The dependable operations are structurally ensured by modeling techniques like FSMs and rule-based logic. Advanced analytics provide intelligent capabilities for prediction

and personalization, while ethical guidelines ensure fairness and protection of rights of the system.

This fusion of components results in an empowered and holistic portfolio of the implementation strategy. It complies technically with the human needs aspects of the senior care. By using it, LGUs can lessen the paper trail and provide better services to the seniors. The system is capable of expanding and modifying to cater to different time needs.

On a larger scale, a profiling system like this aligns with the country's plans for elderly welfare. It is a step towards Philippines' readiness for an increasing elderly population. Through the usage of these researches and records, programmers will be able to formulate ideas retaining practicality, inclusiveness, and sustainability as the pillars. The ultimate aim is the enhancement of the quality of life of senior citizens nationwide.

Significance of the Study

The impact of developing a Centralized Profiling and Record Authentication System for Senior Citizens will be quite significant. It will not only improve public services and digital welfare management but also have an outreach effects on grassroots communities in Pasig City. Integrating Rule-Based Logic and the Finite State Machine (FMS) together, this system is capable of solving actual problems in the delivery of services rather than simply automating data. To this end, the research herein presented will mostly revolve around these groups:

To the Senior Citizens and their Proxies: This system will quite a bit reduce the physical and travel strain of elderly residents and their family members. With the online proxy-token QR

code feature, senior who are bedridden or ill will no longer have to go to the barangay hall physically. Their relatives can also avail of the high priority line at the desk, which will reduced their waiting time and enable them to go home quickly to look after their loved ones.

To the Barangay OSCA Staff: Frontline desk officers will be assisted by a smart digital helper who automatically check data. Rule based logic, which is the fundamental component of the logic, cross checks the submission dates of the documents and the age limits right at the counter (for example the very strict 30 days cut-off for burial assistance). This will help in eliminating human errors, preventing staff from accepting incomplete files and saving personnel from the manual checking of paper intensive folders.

To the Main OSCA Office and LGU Administrators: The system is a transparency and security enabler in the main office of the city hall. With the FSM workflow, administrators are able to know and monitor in real-time the exact location of the batched application folders as they are transferred from the barangay to the main office, so no document will ever be missed or lost. Automated backend validation, such as the 4,000 SSS pension cap for example, also act as safeguards of government funds against double-pension fraud.

To the Commission on Audit (COA) Auditors: A digital footprint that is not only clean but also tamper-proof is automatically generated by the system, thereby matching the corresponding physical documents perfectly. Besides, since the system adheres to the principle of updating the status in separate, consecutive steps, it always offers a transparent history about who did the encoding, checking, and approving each file. Having done this, the LGU should have no problem continuing to adhere to auditing regulations such as COA Circular No. 2021-006.

To Future Researchers and Students of IT/CS: The research could be a manual or a reference for developing localized government systems. In addition, it demonstrates that combining Rule-Based Logic with FSM is far superior and more dependable than heavy AI models on systems that must comply with rigid municipal laws. Students of the future might

also adopt this structure to mechanize other government services like business permits or licensing.

Statement of the Problem

The primary problem of this study is the administrative bottleneck and data disconnect between grassroots Barangay OSCA Help Desks and the Main OSCA Office in Pasig City, where manual document tracking and requirement verification cause severe delays, eligibility assessment errors, and accessibility barriers for low-mobility senior citizens.

Specifically, this study aims to address the following questions:

1. How can frontline barangay desks instantly audit applicant eligibility and local ordinance compliance at the point of intake to prevent human error and incomplete submissions?
2. What mechanisms can be implemented to securely authenticate, prioritize, and process benefit claims for bedridden seniors via proxy without requiring their physical presence?
3. How can a Finite State Machine (FSM) model be utilized to systematically track and validate the real-time status transitions of application batches moving from the barangay to the central office?

4. In what ways can the system enforce localized regulatory parameters, such as the local pension cap and burial assistance deadlines, using a simulated data environment?

Scope and Limitation of the Study

The main purpose of the research is to develop a Centralized Profiling and Record Authentication System for Senior Citizens. To be sure, the system execution is fully complying with the regulations of Pasig City. For the eligibility determination, Rule-Based Logic has been applied whereas the FSM or Finite State Machine has been used to oversee the application process from the barangay to the final city hall approval.

Scope of the Study

- **Target Users and Beneficiaries:** This is mostly meant for senior citizens of Pasig City plus their family members who may be proxies. Besides them, the Barangay OSCA Senior Help Desk Officers/frontline staff, and the Main OSCA Office staff at Tanghalan Pasigueno or Temporary City Hall in Bridgetown, Rosario are considered as the target users too.
- **Core Functions and Modules:**
 - **Online Proxy-Token Generator:** Families who have seniors confined to bed or ill can pre-register through a website module and receive a QR code. When the barangay personnel scan this QR code through a regular webcam, the system queues them in a high-priority digital line.
 - **Pasig Legal Compliance Engine (Rule-Based):** A software module that automatically implements the specific laws of Pasig City. For instance, it verifies the age requirement for Senior IDs (60 years old and above), it sifts

through the SSS pension amount (mandated not to exceed P4,000) about local social pension under Pasig City Ordinance No. 17, Series of 2025, and it determines the absolute 30-working-day time limit for filing the P5,000 Burial Assistance application under Pasig City Ordinance No. 3, Series of 2026.

- **FSM Workflow Tracker:** A background tracking system that updates and observes the changing status of batched applications as they progress from the Barangay draft, to the In Transit, to Under Verification, and finally to Approved for Payroll or Rejected at the Main Office.
- **Hybrid Tracking:** The system handles digital profiles and scanned documents; But, one of the phases entails the physical paperwork. The paper files are collected or bundled at the barangay level and are brought via local government couriers to the Main OSCA Office to comply with the state auditing requirements.

Limitations of the Study

- **Strictly for Pasig City Rules Only:** This system has been designed strictly as the specific ordinances, workflows, and office setups of Pasig City only. It cannot be directly used by other cities or municipalities unless the backend coding and Rule-Based Logic parameters are completely changed to fit their own local laws.
- **No Direct Third-Party Database Integration (Mock API):** Due to stringent data privacy laws and cybersecurity rules that the government has laid down, the system is not allowed to connect live and directly with the external databases of the Social Security System (SSS). Such an operation is only demonstrated through a secure mock database or a simulated data-sharing setup to verify the 4,000 pension restriction.
- **Exclusion of National DSWD Pension Management:** The system itself cannot process applications or grant slots for the National DSWD Indigent Pension program, in fact, it only makes a cross-check with the local database to verify a senior is not a recipient of the national fund, So implementing Pasig's automatic cut-off rule.

- **Offline Barangay Dependencies:** While families have the option to pre-register online and obtain a priority QR code at home, a proxy has to take the responsibility of submitting the hard copies of the documents to the Barangay Help Desk for verification by the staff for audit trail purposes.

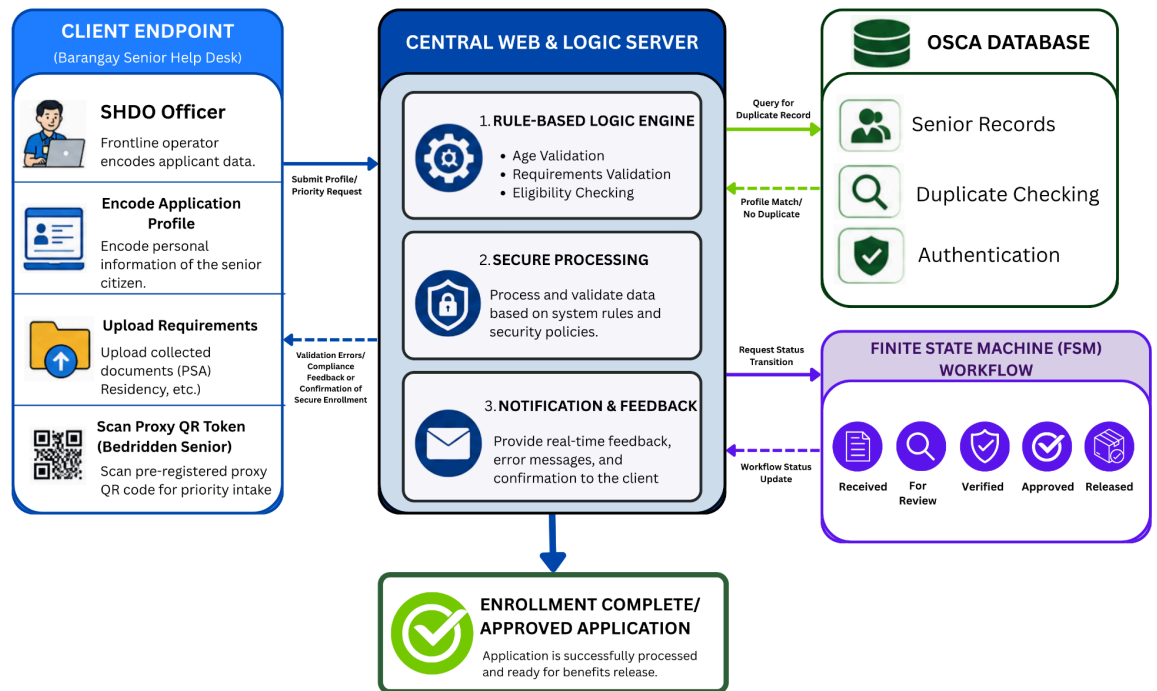
CHAPTER 2

Introduction

This chapter shows the systems architecture, procedures, and development techniques that were used to create, implement, and test the Centralized Profiling and Record Authentication System for Senior Citizens through Rule-Based Logic and Finite State Machine (FSM). Mainly, this chapter aims at giving detailed explanations and programmatic openness on how the research was turned into a system testing phase from the idea stage. Through describing the research design, the working place, the selection of the participants, the making of the system architecture, and the way the results were measured, this chapter works as a complete, step-by-step, duplicable guide to the research method. This not only makes these computational models like the automated rule engines and deterministic state transitions scientifically valid but also legally compliant with the local ordinances of Pasig City and technically meeting the criteria of the public sector.

Research Design

Architectural Framework



Algorithm

Software Engineering (SE) Paradigm

Schematic Diagram

Population and Sampling

Data Collection and Methods

Ethical Considerations

Given that the study entails user feedback collection and public record workflow simulation within the Pasig City LGU, a number of social-ethical considerations were attended to by the researchers to protect all participants and data.

Institutional Consent.By obtaining the officially sanctioned signatures in the permission letters from the OSCA Head and the Barangay Chairpersons before the conduct of the study, the researchers were able to show that the data collection was carried out legally with full administration authorization and without causing any disturbance to the day to day government operations.

Informed Consent and Voluntary Participation.The researchers gave the respondents an Informed Consent Form to read and sign before any testing of the system or distributing the survey was done. The researchers explained the project outline to the participants and gave assurance of their voluntary nature of involvement in the project, and they were given the option to back out of the study at any time without the risk of facing any penalty.

Data Privacy and Confidentiality. (RA 10173 Compliance)In line with the Philippine Data Privacy Act of 2012 (Republic Act No. 10173), no actual, really highly sensitive personal data of senior citizens were revealed or permanently retained during the development stage. The testing was performed through the use of random mock profiles, fictitious SSS numbers, and simulated databases. Besides, the identities and survey ratings of the LGU staff and family proxies who returned the surveys were kept totally confidential, as they were only known by their coded respondent IDs (e.g. Respondent A, Respondent B).

Integrity of Research Data.As regards the reporting of the evaluation results in the development area, the team is committed to maintaining the highest level of academic honesty. No one can accuse them of manipulation, fabrication, or bias as all the survey scores, weighted means, and qualitative remarks taken from the LGU

offices have been properly recorded and accurately computed to reflect the system's genuine final software performance metrics.